

Concept of Abstract Data Types (ADT)

Q. Explain the concept of Abstract Data Type (ADT) with suitable examples. Why is it important in data structures?

> **Definition to Remember**

1. Key Concepts of ADT

- **Abstract** = Implementation details are hidden from the user.

- **Encapsulation:** Data and operations are bundled together.

2. How ADT Works

(Calls: push, pop, peek does NOT know how they work)

ADT Interface Public API (what)

[Hidden Implementation]

3. Classic Example Stack ADT

| Component | Detail |

The user calls `push()` and `pop()` without knowing if the stack is internally implemented using an **array** or a **linked list** that detail is abstracted away.

4. Advantages of ADT

- **Code Reusability:** ADT implementation can be reused across different programs.

- **M

> **Must-Write Points (for 10 marks)**

> 1. AI

> **Quick Recall**

> `AD`
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